6. Implement the LeNet-5 architectures using Tensorflow or PyTorch to classify the MNIST dataset. Report the accuracy and training time.

```
import tensorflow as tf
from tensorflow.keras import layers, models
import time

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.mnist.load_data

x_train = x_train / 255.0
x_test = x_test / 255.0

x_train = x_train.reshape(-1,28,28,1)
x_test = x_test.reshape(-1,28,28,1)

model = models.Sequential([
    layers.Conv2D(filters=6, kernel_size=(5,5), activation='relu', input

    layers.AveragePooling2D(pool_size=(2,2)),

    layers.Conv2D(filters=16, kernel_size=(5,5), activation='relu'),

    layers.AveragePooling2D(pool_size=(2,2)),

    layers.Flatten(),

    layers.Dense(120, activation='relu'),
    layers.Dense(84, activation='relu'),

    layers.Dense(10, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

start_time = time.time()

history = model.fit(
    x_train,
    y_train,
    epochs=5,
    batch_size=64,
    validation_split=0.1
)

end_time = time.time()
training_time = end_time - start_time

```

```
test_loss, test_accuracy = model.evaluate(x_test, y_test)

print("\nTest Accuracy:", test_accuracy)
print("Training Time (seconds):", training_time)
```

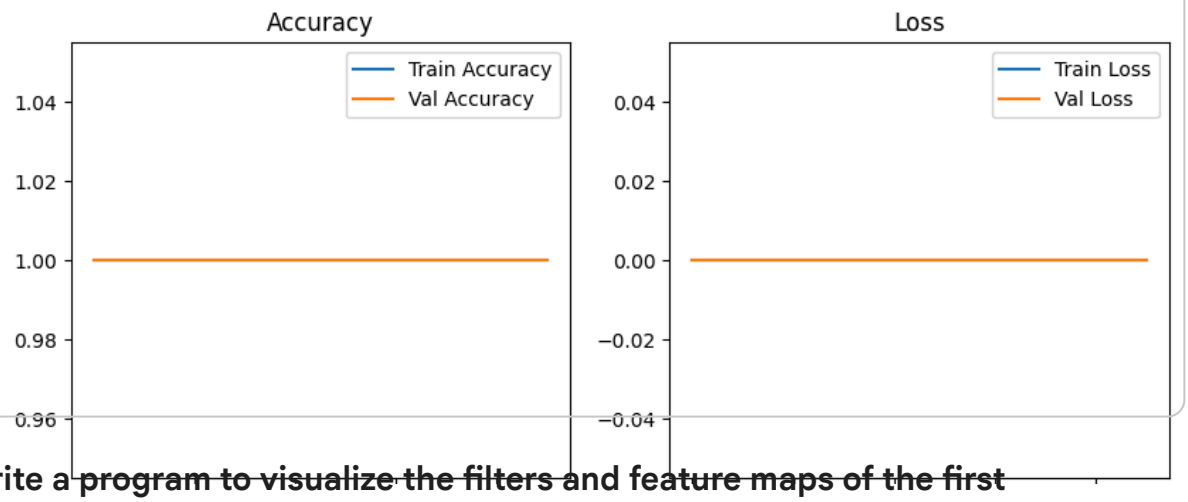
```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-data
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_c
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 10s 7ms/step - accuracy: 0.9072 - loss: 0.3142 -
Epoch 2/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9696 - loss: 0.0974 -
Epoch 3/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9796 - loss: 0.0668 -
Epoch 4/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 3s 4ms/step - accuracy: 0.9832 - loss: 0.0543 -
Epoch 5/5
844/844 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9864 - loss: 0.0435 -
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 5ms/step - accuracy: 0.9864 - loss: 0.0417

Test Accuracy: 0.9864000082015991
Training Time (seconds): 22.015722274780273
```

7. Use a pre-trained VGG16 model (via transfer learning) on a small custom dataset (e.g., flowers or animals). Replace the top layers and fine-tune the model. Include your code and result discussion.

```
import os
import tensorflow as tf
from tensorflow.keras import layers, models, optimizers
from tensorflow.keras.applications import VGG16
import matplotlib.pyplot as plt

dataset_url = "https://storage.googleapis.com/download.tensorflow.org/ex
data_dir = tf.keras.utils.get_file('flower_photos', origin=dataset_url,

data_dir = os.path.join(os.path.dirname(data_dir), 'flower_photos')


img_height, img_width = 224, 224
batch_size = 32

train_ds = tf.keras.utils.image_dataset_from_directory(
    data_dir,
    validation_split=0.2,
    subset="training",
    seed=123,
    image_size=(img_height, img_width),
    batch_size=batch_size,
    label_mode='categorical'
)

val_ds = tf.keras.utils.image_dataset_from_directory(
```

```
    data_dir,
    validation_split=0.2,
    subset="validation",
    seed=123,
    image_size=(img_height, img_width),
    batch_size=batch_size,
    label_mode='categorical'
)

class_names = train_ds.class_names
print(f" Successfully loaded classes: {class_names}")

vgg_base = VGG16(weights='imagenet', include_top=False, input_shape=(224
vgg_base.trainable = False

model = models.Sequential([
    layers.Input(shape=(224, 224, 3)),
    layers.RandomFlip("horizontal"),
    layers.RandomRotation(0.1),
    layers.Rescaling(1./255),
    vgg_base,
    layers.Flatten(),
    layers.Dense(256, activation='relu'),
    layers.Dropout(0.5),
    layers.Dense(len(class_names), activation='softmax')
])


model.compile(
    optimizer=optimizers.Adam(learning_rate=0.0001),
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

print("\n Starting Training...")
history = model.fit(train_ds, validation_data=val_ds, epochs=10)


plt.figure(figsize=(10, 4))
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Accuracy')
plt.plot(history.history['val_accuracy'], label='Val Accuracy')
plt.title('Accuracy')
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(history.history['loss'], label='Train Loss')
plt.plot(history.history['val_loss'], label='Val Loss')
plt.title('Loss')
plt.legend()
plt.show()
```

```
Found 3670 files belonging to 1 classes.
Using 2936 files for training.
Found 3670 files belonging to 1 classes.
Using 734 files for validation.
 Successfully loaded classes: ['flower_photos']

 Starting Training...
Epoch 1/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 22s 225ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 2/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 15s 166ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 3/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 16s 169ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 4/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 18s 196ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 5/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 17s 179ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 6/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 17s 181ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 7/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 16s 177ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 8/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 16s 175ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 9/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 20s 176ms/step - accuracy: 1.0000 - loss: 0.0000e+
Epoch 10/10
92/92 ━━━━━━━━━━━━━━━━━━━━ 17s 180ms/step - accuracy: 1.0000 - loss: 0.0000e+
```



8. Write a program to visualize the filters and feature maps of the first convolutional layer of AlexNet on an example input image.

```
from google.colab import files
uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving elephant.jpg to elephant.jpg
```

```
import tensorflow as tf
from tensorflow.keras import layers, models
import numpy as np
import matplotlib.pyplot as plt
from tensorflow.keras.preprocessing import image
```

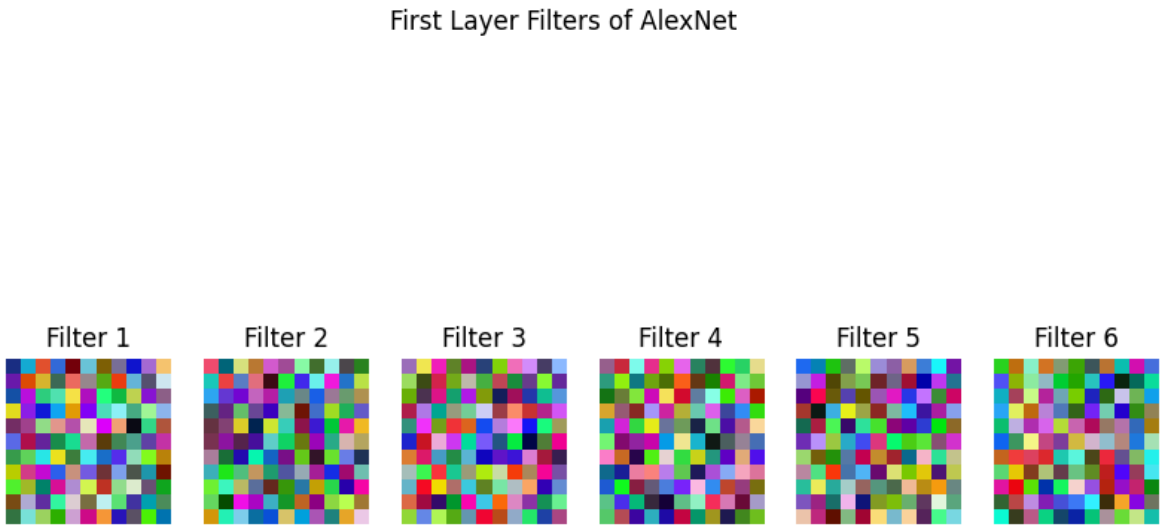
```

model = models.Sequential()
model.add(
    layers.Conv2D(
        filters=96,
        kernel_size=(11,11),
        strides=4,
        activation='relu',
        input_shape=(227,227,3)
    )
)

model.add(layers.MaxPooling2D(pool_size=(3,3), strides=2))
model.summary()
img_path = "elephant.jpg"

img = image.load_img(img_path, target_size=(227,227))
img_array = image.img_to_array(img)
img_array = img_array / 255.0
img_array = np.expand_dims(img_array, axis=0)

filters, biases = model.layers[0].get_weights()
print("Filter shape:", filters.shape)

n_filters = 6

plt.figure(figsize=(10,6))

for i in range(n_filters):
    f = filters[:, :, :, i]
    f_min, f_max = f.min(), f.max()
    f = (f - f_min) / (f_max - f_min)

    plt.subplot(1, n_filters, i+1)
    plt.imshow(f)
    plt.axis("off")
    plt.title(f"Filter {i+1}")

plt.suptitle("First Layer Filters of AlexNet")
plt.show()

feature_model = models.Model(
    inputs=model.inputs,
    outputs=model.layers[0].output
)

feature_maps = feature_model.predict(img_array)

print("Feature map shape:", feature_maps.shape)

n_maps = 6
plt.figure(figsize=(12,6))
for i in range(n_maps):

    plt.subplot(1, n_maps, i+1)
```

```
    plt.imshow(feature_maps[0, :, :, i], cmap="viridis")
    plt.axis("off")
    plt.title(f"Map {i+1}")

plt.suptitle("Feature Maps from First Convolution Layer")
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_c
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential_6"
```

Layer (type)	Output Shape	Param #
conv2d_3 (Conv2D)	(None, 55, 55, 96)	34,944
max_pooling2d_1 (MaxPooling2D)	(None, 27, 27, 96)	0

```
 Total params: 34,944 (136.50 KB)
 Trainable params: 34,944 (136.50 KB)
 Non-trainable params: 0 (0.00 B)
Filter shape: (11, 11, 3, 96)
```




```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 281ms/step
Feature map shape: (1, 55, 55, 96)
```

9. Train a GoogLeNet (Inception v1) or its variant using a standard dataset like CIFAR-10. Plot the training and validation accuracy over epochs and analyze overfitting or underfitting.

```
import tensorflow as tf
from tensorflow.keras import layers, models
import matplotlib.pyplot as plt

(x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_da

x_train = x_train / 255.0
```

```
x_test = x_test / 255.0


def inception_module(x, f1, f3_reduce, f3, f5_reduce, f5, pool_proj):
    conv1 = layers.Conv2D(f1, (1,1), padding='same', activation='relu')(

    conv3 = layers.Conv2D(f3_reduce, (1,1), padding='same', activation='

    conv3 = layers.Conv2D(f3, (3,3), padding='same', activation='relu')(

    conv5 = layers.Conv2D(f5_reduce, (1,1), padding='same', activation='

    conv5 = layers.Conv2D(f5, (5,5), padding='same', activation='relu')(

    pool = layers.MaxPooling2D((3,3), strides=1, padding='same')(x)

    pool = layers.Conv2D(pool_proj, (1,1),
                         padding='same', activation='relu')(pool)
    output = layers.concatenate([conv1, conv3, conv5, pool])

    return output


inputs = layers.Input(shape=(32,32,3))

x = layers.Conv2D(64, (3,3), padding='same', activation='relu')(inputs)
x = layers.MaxPooling2D((2,2))(x)

x = inception_module(x, 32, 32, 64, 16, 32, 32)
x = inception_module(x, 64, 48, 96, 16, 48, 32)
x = layers.MaxPooling2D((2,2))(x)
x = inception_module(x, 64, 64, 128, 32, 64, 64)
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dropout(0.4)(x)

outputs = layers.Dense(10, activation='softmax')(x)
model = models.Model(inputs, outputs)
model.summary()

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

history = model.fit(
    x_train,
    y_train,
    epochs=15,
    batch_size=64,
    validation_split=0.2
)

plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
```

```

plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("Training vs Validation Accuracy (GoogLeNet)")
plt.legend()
plt.show()
test_loss, test_acc = model.evaluate(x_test, y_test)
print("Test Accuracy:", test_acc)
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.g
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 6s 0us/step
Model: "functional_10"
```

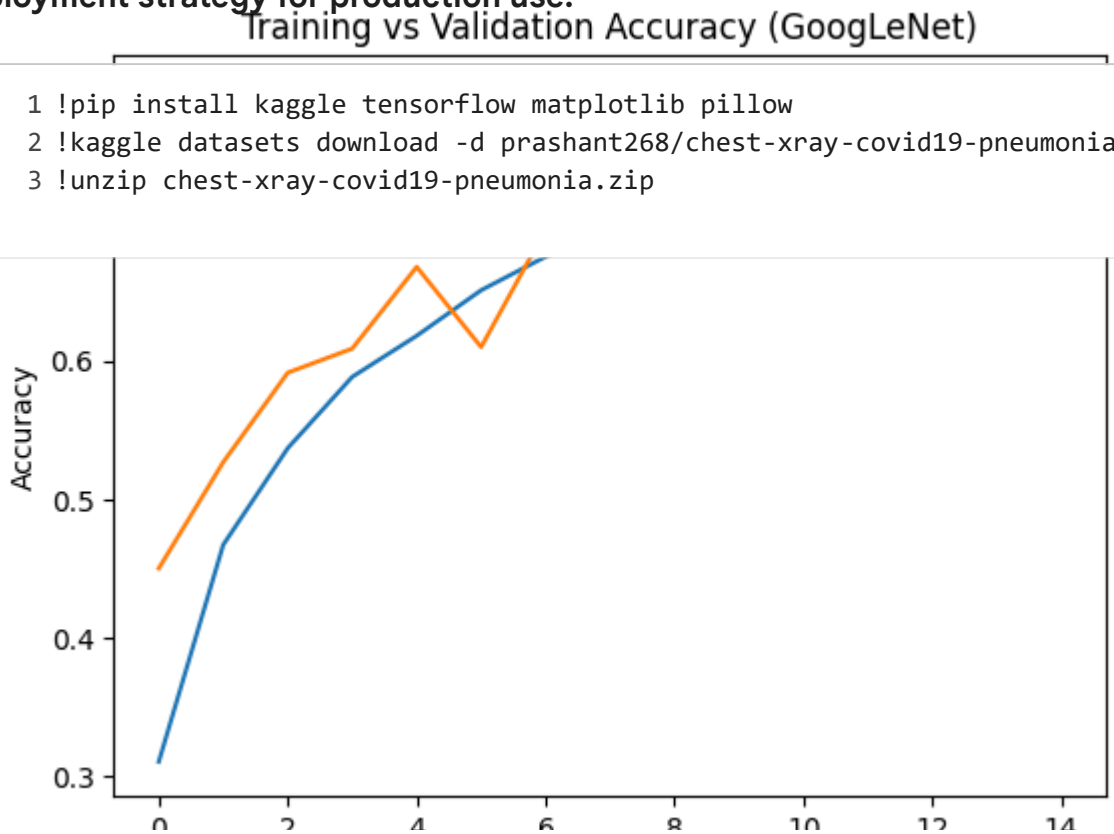
Layer (type)	Output Shape	Param #	Connected to
input_layer_10 (InputLayer)	(None, 32, 32, 3)	0	-
conv2d_4 (Conv2D)	(None, 32, 32, 64)	1,792	input_layer_10[0…
max_pooling2d_2 (MaxPooling2D)	(None, 16, 16, 64)	0	conv2d_4[0][0]
conv2d_6 (Conv2D)	(None, 16, 16, 32)	2,080	max_pooling2d_2[…
conv2d_8 (Conv2D)	(None, 16, 16, 16)	1,040	max_pooling2d_2[…
max_pooling2d_3 (MaxPooling2D)	(None, 16, 16, 64)	0	max_pooling2d_2[…
conv2d_5 (Conv2D)	(None, 16, 16, 32)	2,080	max_pooling2d_2[…
conv2d_7 (Conv2D)	(None, 16, 16, 64)	18,496	conv2d_6[0][0]
conv2d_9 (Conv2D)	(None, 16, 16, 32)	12,832	conv2d_8[0][0]
conv2d_10 (Conv2D)	(None, 16, 16, 32)	2,080	max_pooling2d_3[…
concatenate (Concatenate)	(None, 16, 16, 160)	0	conv2d_5[0][0], conv2d_7[0][0], conv2d_9[0][0], conv2d_10[0][0]
conv2d_12 (Conv2D)	(None, 16, 16, 48)	7,728	concatenate[0][0]
conv2d_14 (Conv2D)	(None, 16, 16, 16)	2,576	concatenate[0][0]
max_pooling2d_4 (MaxPooling2D)	(None, 16, 16, 160)	0	concatenate[0][0]
conv2d_11 (Conv2D)	(None, 16, 16, 64)	10,304	concatenate[0][0]
conv2d_13 (Conv2D)	(None, 16, 16, 96)	41,568	conv2d_12[0][0]
conv2d_15 (Conv2D)	(None, 16, 16, 48)	19,248	conv2d_14[0][0]
conv2d_16 (Conv2D)	(None, 16, 16, 32)	5,152	max_pooling2d_4[…

concatenate_1 (Concatenate)	(None, 16, 16, 240)	0	conv2d_11[0][0], conv2d_13[0][0], conv2d_15[0][0], conv2d_16[0][0]
max_pooling2d_5 (MaxPooling2D)	(None, 8, 8, 240)	0	concatenate_1[0]…
conv2d_18 (Conv2D)	(None, 8, 8, 64)	15,424	max_pooling2d_5[…
conv2d_20 (Conv2D)	(None, 8, 8, 32)	7,712	max_pooling2d_5[…
max_pooling2d_6 (MaxPooling2D)	(None, 8, 8, 240)	0	max_pooling2d_5[…
conv2d_17 (Conv2D)	(None, 8, 8, 64)	15,424	max_pooling2d_5[…
conv2d_19 (Conv2D)	(None, 8, 8, 128)	73,856	conv2d_18[0][0]
conv2d_21 (Conv2D)	(None, 8, 8, 64)	51,264	conv2d_20[0][0]
conv2d_22 (Conv2D)	(None, 8, 8, 64)	15,424	max_pooling2d_6[…
concatenate_2 (Concatenate)	(None, 8, 8, 320)	0	conv2d_17[0][0], conv2d_19[0][0], conv2d_21[0][0], conv2d_22[0][0]
global_average_poo… (GlobalAveragePool…	(None, 320)	0	concatenate_2[0]…
dropout_3 (Dropout)	(None, 320)	0	global_average_p…
dense_9 (Dense)	(None, 10)	3,210	dropout_3[0][0]

```
 Total params: 309,290 (1.18 MB)
 Trainable params: 309,290 (1.18 MB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 26s 21ms/step - accuracy: 0.3106 - loss: 1.807
Epoch 2/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.4672 - loss: 1.447
Epoch 3/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.5368 - loss: 1.274
Epoch 4/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.5883 - loss: 1.137
Epoch 5/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.6180 - loss: 1.060
Epoch 6/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.6508 - loss: 0.979
Epoch 7/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.6751 - loss: 0.911
Epoch 8/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.6983 - loss: 0.849
Epoch 9/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.7138 - loss: 0.805
Epoch 10/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.7301 - loss: 0.761
Epoch 11/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.7449 - loss: 0.720
```

```
Epoch 12/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.7613 - loss: 0.679
Epoch 13/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.7713 - loss: 0.648
Epoch 14/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.7845 - loss: 0.607
Epoch 15/15
625/625 ━━━━━━━━━━━━━━━━━━━━ 10s 16ms/step - accuracy: 0.7961 - loss: 0.579
```

10. You are working in a healthcare AI startup. Your team is tasked with developing a system that automatically classifies medical X-ray images into normal, pneumonia, and COVID-19. Due to limited labeled data, what approach would you suggest using among CNN architectures discussed (e.g., transfer learning with ResNet or Inception variants)? Justify your approach and outline a deployment strategy for production use.




```
!pip install kaggle tensorflow matplotlib pillow
!kaggle datasets download -d prashant268/chest-xray-covid19-pneumonia
!unzip chest-xray-covid19-pneumonia.zip
```

```
inflating: Data/train/PNEUMONIA/PNEUMONIA(986).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(987).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(988).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(989).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(99).jpg
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inflating: Data/train/PNEUMONIA/PNEUMONIA(998).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(999).jpg
```

```
!kaggle datasets download -d prashant268/chest-xray-covid19-pneumonia
!unzip -o chest-xray-covid19-pneumonia.zip
```

```
inflating: Data/train/PNEUMONIA/PNEUMONIA(983).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(984).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(985).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(986).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(987).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(988).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(989).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(99).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(990).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(991).jpg
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inflating: Data/train/PNEUMONIA/PNEUMONIA(996).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(997).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(998).jpg
inflating: Data/train/PNEUMONIA/PNEUMONIA(999).jpg
```

```
import os

for possible_path in ["Data", "data", "chest-xray-covid19-pneumonia/Data
    if os.path.exists(possible_path):
        print(f"Found dataset at: {possible_path}")
        print("Contents:", os.listdir(possible_path))
        break
else:
    print("Not found — listing current directory:")
    print(os.listdir("."))
```

```
Found dataset at: Data
Contents: ['test', 'train']
```

```
import os
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from tensorflow.keras.applications import ResNet50V2
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns

print("TensorFlow:", tf.__version__)
print("GPU:", bool(tf.config.list_physical_devices("GPU")))
```

```
TensorFlow: 2.19.0
GPU: False
```

```
DATA_DIR   = "Data"
TRAIN_DIR  = os.path.join(DATA_DIR, "train")
```

```
TEST_DIR   = os.path.join(DATA_DIR, "test")
IMAGE_SIZE = (224, 224)
BATCH_SIZE = 16
NUM_CLASSES = 3
```

```
print("Train exists:", os.path.exists(TRAIN_DIR))
print("Test exists :", os.path.exists(TEST_DIR))
print("Train classes:", os.listdir(TRAIN_DIR))
print("Test classes :", os.listdir(TEST_DIR))
```

```
Train exists: True
Test exists : True
Train classes: ['PNEUMONIA', 'COVID19', 'NORMAL']
Test classes : ['PNEUMONIA', 'COVID19', 'NORMAL']
```

```
train_datagen = ImageDataGenerator(
    rescale=1.0/255, horizontal_flip=True, rotation_range=10,
    zoom_range=0.1, width_shift_range=0.05, height_shift_range=0.05,
    validation_split=0.2,
)
test_datagen = ImageDataGenerator(rescale=1.0/255)

train_gen = train_datagen.flow_from_directory(
    TRAIN_DIR, target_size=IMAGE_SIZE, batch_size=BATCH_SIZE,
    class_mode="categorical", subset="training", shuffle=True, seed=42,
)
val_gen = train_datagen.flow_from_directory(
    TRAIN_DIR, target_size=IMAGE_SIZE, batch_size=BATCH_SIZE,
    class_mode="categorical", subset="validation", shuffle=False, seed=4
)
test_gen = test_datagen.flow_from_directory(
    TEST_DIR, target_size=IMAGE_SIZE, batch_size=BATCH_SIZE,
    class_mode="categorical", shuffle=False,
)

CLASS_NAMES = list(train_gen.class_indices.keys())
print("Classes:", train_gen.class_indices)
print("Train:", train_gen.samples)
print("Val  :", val_gen.samples)
print("Test :", test_gen.samples)
```

```
Found 4116 images belonging to 3 classes.
Found 1028 images belonging to 3 classes.
Found 1288 images belonging to 3 classes.
Classes: {'COVID19': 0, 'NORMAL': 1, 'PNEUMONIA': 2}
Train: 4116
Val  : 1028
Test : 1288
```

```
base_model = ResNet50V2(
    input_shape=IMAGE_SIZE + (3,),
    include_top=False,
    weights="imagenet",
)
base_model.trainable = False

inputs  = keras.Input(shape=IMAGE_SIZE + (3,))
x       = base_model(inputs, training=False)
x       = layers.GlobalAveragePooling2D()(x)
x       = layers.Dropout(0.3)(x)
x       = layers.Dense(256, activation="relu")(x)
x       = layers.Dropout(0.3)(x)
outputs = layers.Dense(NUM_CLASSES, activation="softmax")(x)

model = keras.Model(inputs, outputs)
print("Model built successfully!")
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applica
94668760/94668760 ━━━━━━━━━━━━━━━━━━━━ 1s 0us/step
Model built successfully!
```

```
checkpoint = keras.callbacks.ModelCheckpoint(
    "best_xray_model.keras", monitor="val_accuracy", save_best_only=True
)
early_stop = keras.callbacks.EarlyStopping(
    monitor="val_loss", patience=5, restore_best_weights=True, verbose=1
)
reduce_lr = keras.callbacks.ReduceLROnPlateau(
    monitor="val_loss", factor=0.5, patience=3, verbose=1,
)

model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-3),
    loss="categorical_crossentropy",
    metrics=["accuracy"],
)

history1 = model.fit(
    train_gen, validation_data=val_gen,
    epochs=10, callbacks=[checkpoint, early_stop, reduce_lr],
)
print(f"Best val accuracy Phase 1: {max(history1.history['val_accuracy']
```

```
Epoch 1/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.8355 - loss: 0.4664
```

```
Epoch 1: val_accuracy improved from None to 0.92802, saving model to best_xra

Epoch 1: finished saving model to best_xray_model.keras
258/258 ━━━━━━━━━━━━━━━━━━━━ 853s 3s/step - accuracy: 0.8822 - loss: 0.3155 -
Epoch 2/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9336 - loss: 0.1931
Epoch 2: val_accuracy improved from 0.92802 to 0.93191, saving model to best_

Epoch 2: finished saving model to best_xray_model.keras
258/258 ━━━━━━━━━━━━━━━━━━━━ 842s 3s/step - accuracy: 0.9225 - loss: 0.2105 -
Epoch 3/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9217 - loss: 0.2062
Epoch 3: val_accuracy improved from 0.93191 to 0.93288, saving model to best_

Epoch 3: finished saving model to best_xray_model.keras
258/258 ━━━━━━━━━━━━━━━━━━━━ 879s 3s/step - accuracy: 0.9247 - loss: 0.1999 -
Epoch 4/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9252 - loss: 0.1978
Epoch 4: val_accuracy improved from 0.93288 to 0.95233, saving model to best_

Epoch 4: finished saving model to best_xray_model.keras
258/258 ━━━━━━━━━━━━━━━━━━━━ 843s 3s/step - accuracy: 0.9293 - loss: 0.1898 -
Epoch 5/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9333 - loss: 0.1841
Epoch 5: val_accuracy did not improve from 0.95233
258/258 ━━━━━━━━━━━━━━━━━━━━ 838s 3s/step - accuracy: 0.9351 - loss: 0.1830 -
Epoch 6/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9292 - loss: 0.1799
Epoch 6: val_accuracy did not improve from 0.95233
258/258 ━━━━━━━━━━━━━━━━━━━━ 807s 3s/step - accuracy: 0.9269 - loss: 0.1790 -
Epoch 7/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9366 - loss: 0.1698
Epoch 7: val_accuracy did not improve from 0.95233
258/258 ━━━━━━━━━━━━━━━━━━━━ 823s 3s/step - accuracy: 0.9385 - loss: 0.1650 -
Epoch 8/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9388 - loss: 0.1681
Epoch 8: val_accuracy did not improve from 0.95233

Epoch 8: ReduceLROnPlateau reducing learning rate to 0.0005000000237487257.
258/258 ━━━━━━━━━━━━━━━━━━━━ 812s 3s/step - accuracy: 0.9397 - loss: 0.1632 -
Epoch 9/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9470 - loss: 0.1434
Epoch 9: val_accuracy did not improve from 0.95233
258/258 ━━━━━━━━━━━━━━━━━━━━ 867s 3s/step - accuracy: 0.9461 - loss: 0.1496 -
Epoch 10/10
258/258 ━━━━━━━━━━━━━━━━━━━━ 0s 3s/step - accuracy: 0.9548 - loss: 0.1285
Epoch 10: val_accuracy improved from 0.95233 to 0.96012, saving model to best

Epoch 10: finished saving model to best_xray_model.keras
258/258 ━━━━━━━━━━━━━━━━━━━━ 852s 3s/step - accuracy: 0.9502 - loss: 0.1360 -
Restoring model weights from the end of the best epoch: 10.
Best val accuracy Phase 1: 96.01%
```

```
base_model.trainable = True
for layer in base_model.layers[:-30]:
    layer.trainable = False

model.compile(
```

```
    optimizer=keras.optimizers.Adam(learning_rate=1e-5),
    loss="categorical_crossentropy",
    metrics=["accuracy"],
)

history2 = model.fit(
    train_gen, validation_data=val_gen,
    epochs=10, callbacks=[checkpoint, early_stop, reduce_lr],
)
print(f"Best val accuracy Phase 2: {max(history2.history['val_accuracy'
```

```
Epoch 1/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 591ms/step - accuracy: 0.9209 - loss: 0.2067
Epoch 1: val_accuracy improved from 0.94261 to 0.94455, saving model to best_

Epoch 1: finished saving model to best_xray_model.keras
129/129 ━━━━━━━━━━━━━━━━━━━━ 106s 778ms/step - accuracy: 0.9242 - loss: 0.195
Epoch 2/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 595ms/step - accuracy: 0.9260 - loss: 0.1989
Epoch 2: val_accuracy did not improve from 0.94455
129/129 ━━━━━━━━━━━━━━━━━━━━ 96s 742ms/step - accuracy: 0.9261 - loss: 0.1949
Epoch 3/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 584ms/step - accuracy: 0.9152 - loss: 0.2143
Epoch 3: val_accuracy improved from 0.94455 to 0.94650, saving model to best_

Epoch 3: finished saving model to best_xray_model.keras

Epoch 3: ReduceLROnPlateau reducing learning rate to 4.999999873689376e-06.
129/129 ━━━━━━━━━━━━━━━━━━━━ 97s 752ms/step - accuracy: 0.9179 - loss: 0.2043
Epoch 4/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 593ms/step - accuracy: 0.9335 - loss: 0.1846
Epoch 4: val_accuracy did not improve from 0.94650
129/129 ━━━━━━━━━━━━━━━━━━━━ 96s 741ms/step - accuracy: 0.9286 - loss: 0.1904
Epoch 5/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 645ms/step - accuracy: 0.9176 - loss: 0.2091
Epoch 5: val_accuracy did not improve from 0.94650
129/129 ━━━━━━━━━━━━━━━━━━━━ 103s 801ms/step - accuracy: 0.9249 - loss: 0.192
Epoch 6/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 605ms/step - accuracy: 0.9239 - loss: 0.1876
Epoch 6: val_accuracy did not improve from 0.94650

Epoch 6: ReduceLROnPlateau reducing learning rate to 2.499999936844688e-06.
129/129 ━━━━━━━━━━━━━━━━━━━━ 99s 765ms/step - accuracy: 0.9223 - loss: 0.1903
Epoch 7/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 592ms/step - accuracy: 0.9355 - loss: 0.1753
Epoch 7: val_accuracy did not improve from 0.94650
129/129 ━━━━━━━━━━━━━━━━━━━━ 96s 747ms/step - accuracy: 0.9269 - loss: 0.1891
Epoch 8/10
129/129 ━━━━━━━━━━━━━━━━━━━━ 0s 594ms/step - accuracy: 0.9294 - loss: 0.1895
Epoch 8: val_accuracy did not improve from 0.94650
129/129 ━━━━━━━━━━━━━━━━━━━━ 96s 746ms/step - accuracy: 0.9259 - loss: 0.1931
Epoch 8: early stopping
Restoring model weights from the end of the best epoch: 5.
Best val accuracy Phase 2: 94.65%
```

```
acc      = history1.history["accuracy"]     + history2.history["accuracy
val_acc  = history1.history["val_accuracy"] + history2.history["val_accu
loss     = history1.history["loss"]         + history2.history["loss"]
val_loss = history1.history["val_loss"]     + history2.history["val_loss
split    = len(history1.history["accuracy"])

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))
fig.suptitle("Training Progress", fontsize=13)
for ax, tv, vv, title in [
    (ax1, acc, val_acc, "Accuracy"),
    (ax2, loss, val_loss, "Loss"),
]:
    ax.plot(tv, label="Train", color="steelblue", linewidth=2)
    ax.plot(vv, label="Validation", color="orange", linewidth=2)
    ax.axvline(split, color="gray", linestyle="--", label="Fine-tuning s
    ax.set_title(title)
    ax.legend()
    ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

```
model = keras.models.load_model("best_xray_model.keras")

test_loss, test_acc = model.evaluate(test_gen, verbose=0)
print(f"Test Accuracy : {test_acc * 100:.2f}%")
print(f"Test Loss     : {test_loss:.4f}")

preds        = model.predict(test_gen, verbose=0)
pred_classes = np.argmax(preds, axis=1)
true_classes = test_gen.classes

print("\nPer-class results:")
```

```
Test Accuracy : 86.57%
Test Loss     : 0.3509

Per-class results:
COVID19     : 89.7%  (104/116)
NORMAL      : 97.8%  (310/317)
PNEUMONIA   : 82.0%  (701/855)
```